

Yachi Darji

darjiyachi8@gmail.com | +1 (872) 899-4600 | [linkedin.com/in/yachi-darji](https://www.linkedin.com/in/yachi-darji) | github.com/yachi2605 | yachidarji.com

EDUCATION

Illinois Institute of Technology, Chicago, IL - Master of Science, Data Science, GPA: 3.72/4.0 May 2026

Gujarat Technological University, Gujarat, India - Bachelor of Engineering, Information Technology, GPA: 3.8/4.0 May 2024

PROFESSIONAL EXPERIENCE

Data Scientist, Labelmaster (Co-op) | Chicago, IL Jan 2026 - May 2026

- Engineered multi-horizon **LSTM** forecasting platform with 14 derived features and direct multi-output modeling, reducing WAPE from 46% to 31.5% across six business units supporting \$2M+ quarterly procurement planning.
- Established rolling-origin backtesting framework with 52-week validation windows and **VIF**-based feature selection, evaluating 100+ model iterations and reducing forecast bias by 23%.
- Automated enterprise demand forecasting pipelines using **AWS CloudWatch**, **SNS**, and scheduled retraining, generating SLA compliant weekly forecasts for 50K+ inventory records while eliminating 10+ analyst hours weekly.

Data Scientist, August Infotech | Gujarat, India Jan 2024 - Jul 2024

- Built **XGBoost** churn prediction model using 200+ behavioral and transactional features, elevating **AUC-ROC** from 0.68 to 0.81 while supporting retention campaigns across high-risk customer segments.
- Spearheaded experimentation framework integrating Bayesian sequential analysis and **CUPED** variance reduction, accelerating experiment cycles by 40% and shortening feature-release decision timelines.
- Analyzed six-stage conversion funnel using **SQL**, **cohort analysis**, and behavioral segmentation, uncovering conversion bottlenecks that lifted trial-to-paid conversions by 9.4%.

Machine Learning Intern, Orion Technolabs | Gujarat, India Jun 2022 - May 2023

- Implemented lead-scoring platform evaluating **Logistic Regression**, **Random Forest**, **XGBoost**, and **MLP** models, achieving 84% precision while prioritizing high-value leads generating 65% of closed conversions.
- Conducted **RFM** segmentation across 50K+ customer records using **K-Means clustering**, identifying five buyer segments that strengthened campaign engagement and response rates.
- Deployed a **Flask REST API** integrating feature pipelines for real-time lead scoring, reducing processing latency from two hours to under five seconds per batch.

PROJECTS

SupplyChainIQ | AI-Powered Supply Chain Forecasting Platform [Link](#)

- Architected a production-grade demand forecasting platform (FastAPI · React · PostgreSQL) generating 30-day SKU-level forecasts with confidence intervals, deployed on Vercel and Render with multi-tenant API key authentication.
- Formulated a model selection pipeline to backtest SeasonalNaive7, Prophet, and XGBoost; resolved MAPE unreliability on low-volume products using composite ranking (MAE 40% · RMSE 40% · MAPE 20%), improving model selection accuracy.
- Developed schema-agnostic CSV ingestion layer using intent-based keyword scoring to auto-map any column layout to internal schema eliminating manual data prep and enabling plug-and-play onboarding for non-technical users.
- Delivered end-to-end supply chain intelligence tool raw sales CSV to ML forecast, prioritized restock recommendations with urgency flags, Monte Carlo supplier disruption simulation, and exportable PDF manager report.

RentPilot | AI Renter Safety Platform | Python · FastAPI · OpenAI API · RAG · Next.js · Supabase · Vercel [Link](#)

- Designed a production RAG pipeline for tenant rights Q&A, retrieving statutes from a curated PostgreSQL knowledge base, injecting grounding context, and enforcing model refusals for out-of-scope queries, reducing hallucination risk without requiring a vector database.
- Built a multi-stage LLM inference pipeline for unstructured lease PDF analysis implemented text extraction, PII redaction, and prompt sanitization before model invocation, returning structured risk scores and clause-level red flags via OpenAI function calling.
- Reduced AI inference costs by implementing SHA-256 content-hash caching at the API layer identical queries return cached results instantly; combined with per-user quota enforcement to build a cost-bounded, multi-tenant AI backend.
- Shipped anonymous-first AI tooling used without authentication architected optional JWT middleware and localStorage-based result persistence so users get immediate value before signing up, improving conversion funnel design

SKILLS

Programming Language : Python, SQL, PySpark, C++, Bash

Machine Learning & AI : Scikit-learn, XGBoost, LightGBM, Logistic Regression, Random Forest, K-Means Clustering, PyTorch, TensorFlow, Keras, LSTM, Transformers, Hugging Face

Statistics & Analytics : A/B Testing, Bayesian Statistics, CUPED, Propensity Score Matching, Cohort Analysis, Time Series Forecasting, Hypothesis Testing, Power Analysis, Feature Engineering, Feature Selection, Model Validation, Exploratory Data Analysis (EDA)

Data Engineering & Deployment : ETL Pipelines, Data Modeling, Flask, FastAPI, REST APIs, API Development, Streamlit

MLOps & Cloud : AWS (SageMaker, EC2, S3, Lambda, ECR, CloudWatch, IAM), MLflow, Docker, Airflow, CI/CD, Model Monitoring

Tools & Visualization : Git, GitHub, Linux, Jupyter Notebook, Tableau, Power BI, Matplotlib, Seaborn